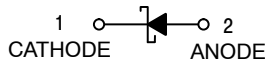
Schottky Barrier Diode

NSR0240MX2

40 V SCHOTTKY BARRIER DIODE

Schottky barrier diodes are optimized for very low forward voltage drop and low leakage current and are used in a wide range of dc-dc converter, clamping and protection applications in portable devices. NSR0240MX2 in the X2DFN2 miniature package enables designers to meet the challenging task of achieving higher efficiency and meeting reduced space requirements.



Features

- Very Low Forward Voltage Drop: 460 mV @ 100 mA
- Low Reverse Current: 0.2 μA @ 25 V V_R
- 200 mA of Continuous Forward Current
- Very High Switching Speed
- Low Capacitance: CT = 7 pF
- X2DFNW2 Wettable Flank Package for Optimal Automated Optical Inspection (AOI)
- NSV prefix PN for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable
- These Devices are Pb-Free, Halogen Free/BFR Free and are RoHS Compliant

Typical Applications

- LCD and Keypad Backlighting
- Camera Photo Flash
- Buck and Boost dc-dc Converters
- Reverse Voltage and Current Protection
- Clamping & Protection

Markets

- Mobile Handsets & Notebook PCs
- Automotive Electronic Control Units
- GPS

MARKING DIAGRAMS



X2DFN2
CASE 714AB





X2DFNW2
CASE 711BG



R = Specific Device Code
M = Month Code

ORDERING INFORMATION

Device	Package	Shipping†
NSR0240MX2T5G	X2DFN2 (Pb-Free)	8,000 / Tape & Reel
NSR0240MX2WT5G	X2DFNW2 (Pb-Free)	
NSVR0240MX2WT5G		

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

MAXIMUM RATINGS

Symbol	Rating	Value	Unit
V_R	Reverse Voltage	40	V
I_F	Forward Current (DC)	200	mA
I_{FSM}	Non-Repetitive Peak Forward Surge Current, Square Wave, 10 ms	3.0	A
I_{FRM}	Repetitive Peak Forward Current, Square Wave, 1.0 ms, D.C. = 25%	1.0	A
ESD	ESD Rating: Human Body Model Machine Model	Class 1C Class A	

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

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THERMAL CHARACTERISTICS

Symbol	Characteristic	Min	Typ	Max	Unit
$R_{\theta JA}$ P_D	Thermal Resistance Junction-to-Ambient (Note 1) Total Power Dissipation @ T_A = 25 °C	–	–	 400 300	 °C/W mW
T_J, T_{stg}	Junction and Storage Temperature Range	–	–	–55 to +150	°C

1. FR-4, 20 mm^2, 1 oz. Cu.

ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)

Symbol	Characteristic	Min	Typ	Max	Unit
I_R	Reverse Leakage (V_R = 25 V) (V_R = 40 V)	–	 0.2 0.8	 0.55 5.0	µA
V_F	Forward Voltage (I_F = 0.1 mA) (I_F = 1.0 mA) (I_F = 10 mA) (I_F = 100 mA) NSVR0240MX2WT5G NSVR0240MX2WT5G (T_A = 125 °C) (I_F = 200 mA) NSVR0240MX2WT5G	–	 0.21 0.27 0.34 0.46 0.49 0.39 0.54 0.60	 0.24 0.30 0.365 0.50 0.53 0.50 0.60 0.65	V
CT	Total Capacitance (V_R = 1.0 V, f = 1 MHz)	–	7.0	–	pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

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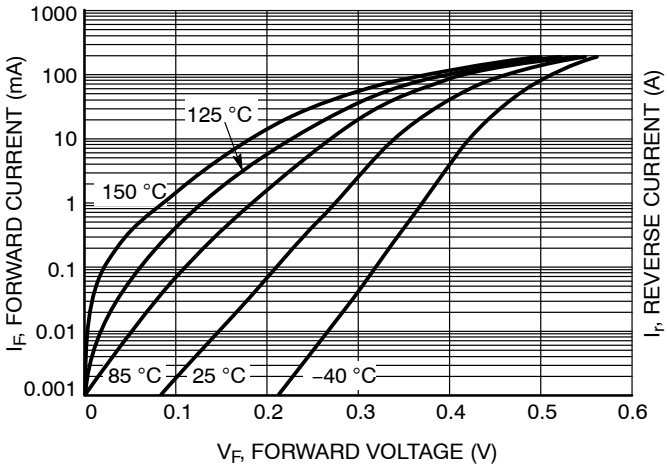


Figure 1. Forward Voltage

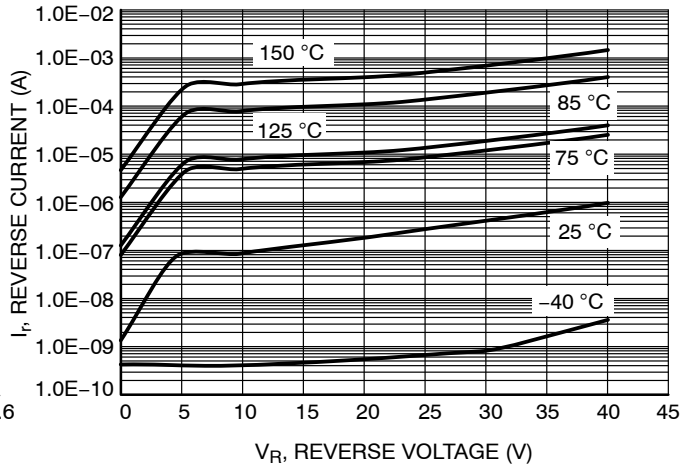


Figure 2. Leakage Current

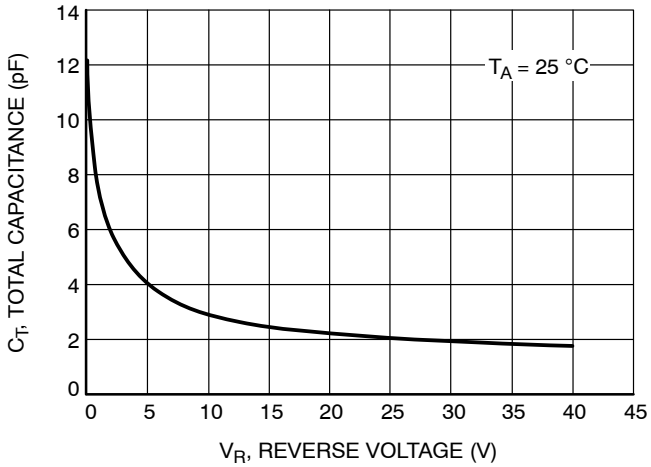


Figure 3. Total Capacitance

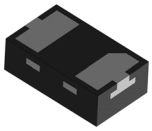
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REVISION HISTORY

Revision	Description of Changes	Date
5	Update of the front page 711BG case outline 3D model.	11/19/2025

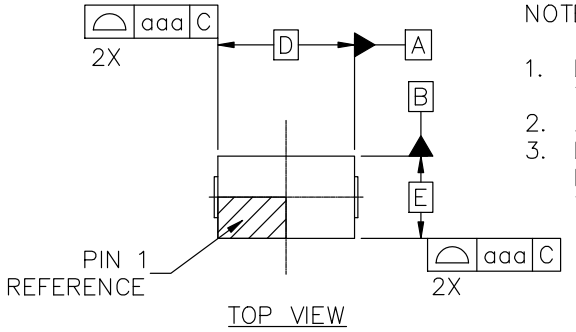
This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

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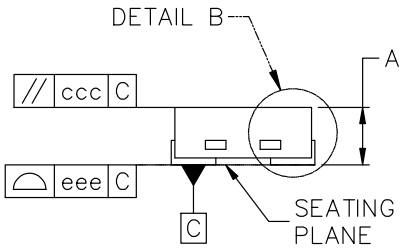


X2DFNW-2 1.00x0.60x0.37, 0.65P
CASE 711BG
ISSUE F

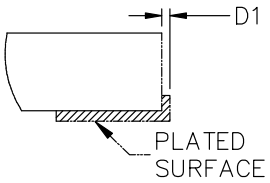
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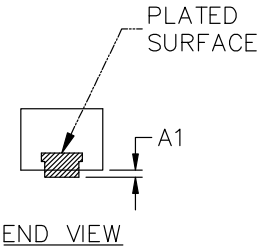
TOP VIEW



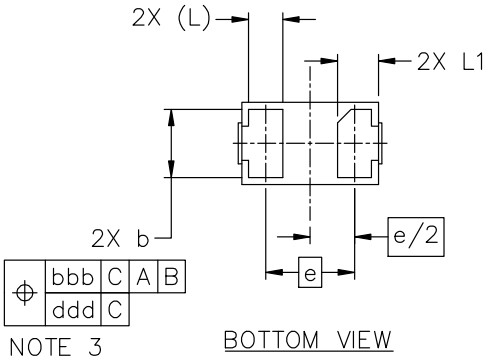
SIDE VIEW



DETAIL "B"
SCALE 2:1



END VIEW



BOTTOM VIEW

NOTES:

1. DIMENSIONING AND TOLERANCING CONFORM TO ASME Y14.5−2018.
2. ALL DIMENSION ARE IN MILLIMETERS.
3. DIMENSION b APPLIES TO THE PLATED TERMINAL AND IS MEASURED BETWEEN 0.15 AND 0.30 FROM THE TERMINAL TIP.

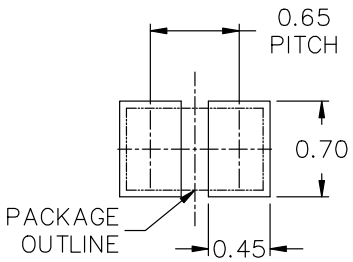
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A	0.34	0.37	0.40
A1	~	~	0.05
b	0.45	0.50	0.55
D		1.00 BSC	
D1	~	~	0.05
E		0.60 BSC	
e		0.65 BSC	
L		0.22 REF	
L1	0.24	0.28	0.34
TOLERANCE FORM & POSITION			
aaa		0.05	
bbb		0.10	
ccc		0.05	
ddd		0.05	
eee		0.05	

GENERIC MARKING DIAGRAM*



XX = Specific Device Code
M = Date Code

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G", may or not be present. Some products may not follow the Generic Marking.



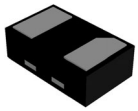
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* FOR ADDITIONAL INFORMATION ON OUR Pb−FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE onsemi SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D.

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DESCRIPTION:	**X2DFNW-2 1.00x0.60x0.37, 0.65P**		**PAGE 1 OF 1**

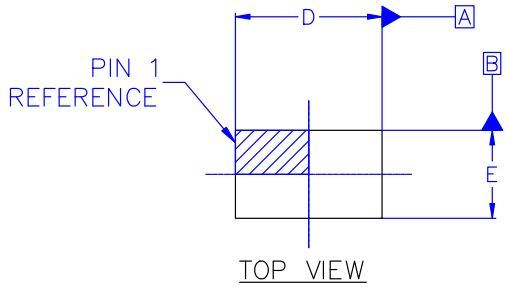
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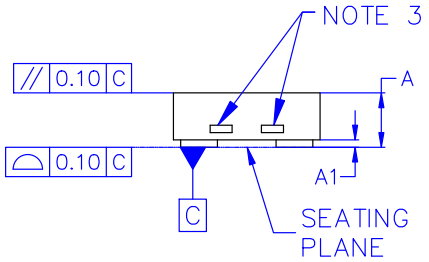


X2DFN−2 1.00x0.60x0.37, 0.65P
CASE 714AB
ISSUE D

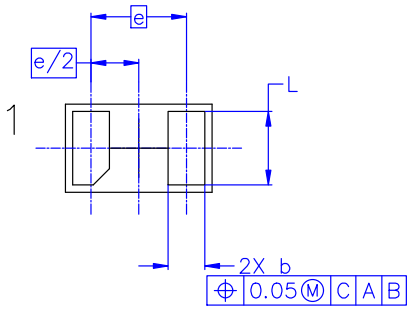
DATE 25 SEP 2026



TOP VIEW



SIDE VIEW

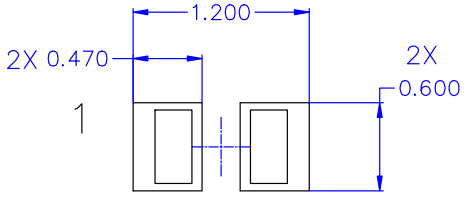


BOTTOM VIEW

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2. ALL DIMENSION ARE IN MILLIMETERS.
3. EXPOSED COPPER ALLOWED AS SHOW.

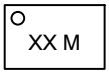
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	MIN.	NOM.	MAX.
A	0.34	0.37	0.40
A1	−−−	0.03	0.050
b	0.20	0.25	0.30
D	0.95	1.00	1.05
E	0.55	0.60	0.65
e	0.65 BSC		
L	0.45	0.50	0.55



RECOMMENDED MOUNTING FOOTPRINT*

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GENERIC
MARKING DIAGRAM*



XX = Specific Device Code
M = Date Code

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “ ▪”, may or may not be present. Some products may not follow the Generic Marking.

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